

Literature-implied answer for arXiv:2310.01450

The Continuous Kraus Conjecture follows from POVM entropy

Status. literature_implied_answer (full). The central inequality is a rank-one continuous-frame specialization of Rumin's 2011 entropic uncertainty principle for positive operator-valued measures (POVMs) [2]. The identification below is not presented as a new core uncertainty theorem.

Source question. Let (Ω, μ) and (Δ, ν) be measure spaces, and let $\{\tau_\alpha\}_{\alpha \in \Omega}$ and $\{\omega_\beta\}_{\beta \in \Delta}$ be 1-bounded continuous Parseval frames for a Hilbert space $\mathcal{H}$. With

$$c = \sup_{\alpha \in \Omega, \beta \in \Delta} |\langle \tau_\alpha, \omega_\beta \rangle|,$$

Conjecture 2.5 on page 4 of Krishna [1] asks whether

$$S_\tau(h) + S_\omega(h) \geq -2 \log c \quad (h \in \mathcal{H}_\tau \cap \mathcal{H}_\omega). \quad (\text{CK})$$

Decisive supporting theorem. Rumin's Theorem 1.1 on page 2 of [2] associates probability densities to two POVMs P, Q . If their Liouville measure satisfies $\mu_{PQ} \leq \mu_P \otimes \mu_Q$, then the sum of the two relative entropies is at least the von Neumann entropy of the state. For a pure state, the latter entropy is zero.

Identification in the sigma-finite case. The frames define POVMs by

$$\begin{aligned} \langle P(E)x, y \rangle &= \int_E \langle x, \tau_\alpha \rangle \langle \tau_\alpha, y \rangle d\mu(\alpha), \\ \langle Q(F)x, y \rangle &= \int_F \langle x, \omega_\beta \rangle \langle \omega_\beta, y \rangle d\nu(\beta). \end{aligned}$$

Parseval gives $P(\Omega) = Q(\Delta) = I$. For the pure state $\rho_h = |h\rangle\langle h|$, where $\|h\| = 1$, put

$$p(\alpha) = |\langle h, \tau_\alpha \rangle|^2, \quad q(\beta) = |\langle h, \omega_\beta \rangle|^2.$$

These are the two measurement densities, and the Liouville density is

$$|\langle \tau_\alpha, \omega_\beta \rangle|^2 \leq c^2.$$

Use the reference measures $\mu_P = c\mu$ and $\mu_Q = c\nu$. Their product dominates the Liouville measure, while the measurement densities with respect to them are p/c and q/c . Rumin's theorem therefore yields

$$\begin{aligned} 0 &\leq - \int_\Omega \frac{p}{c} \log\left(\frac{p}{c}\right) c d\mu - \int_\Delta \frac{q}{c} \log\left(\frac{q}{c}\right) c d\nu \\ &= S_\tau(h) + S_\omega(h) + 2 \log c, \end{aligned}$$

which is exactly (CK).

The unrestricted measure-space wording. Rumin assumes sigma-finite reference measures in order to obtain abstract Radon–Nikodym densities. Here the densities p, q are already explicit, and his proof specializes without that step. For completeness, assume the two entropies are finite (otherwise (CK) is immediate) and define

$$A = \frac{1}{c} \int_{\Omega} p(\alpha), |\tau_{\alpha}\rangle\langle\tau_{\alpha}| d\mu(\alpha), \quad B = \frac{1}{c} \int_{\Delta} q(\beta), |\omega_{\beta}\rangle\langle\omega_{\beta}| d\nu(\beta).$$

These positive operators are trace class because the frames are 1-bounded and $\int p = \int q = 1$. Moreover, using an iterated positive integral,

$$\begin{aligned} \text{Tr}(A^{1/2} B A^{1/2}) &= \frac{1}{c^2} \int_{\Omega} p(\alpha) \left(\int_{\Delta} q(\beta) |\langle\tau_{\alpha}, \omega_{\beta}\rangle|^2, d\nu(\beta) \right) d\mu(\alpha) \\ &\leq \left(\int_{\Omega} p d\mu \right) \left(\int_{\Delta} q d\nu \right) = 1. \end{aligned}$$

Golden–Thompson and the Gibbs variational formula, exactly as in the proof of Rumin’s Theorem 1.1, give

$$-\langle(\log A)h, h\rangle - \langle(\log B)h, h\rangle \geq 0.$$

Operator Jensen for the operator-convex function $-\log$, applied to the two POVM integrals, gives

$$\begin{aligned} -\langle(\log A)h, h\rangle &\leq -\int_{\Omega} p \log(p/c) d\mu = S_{\tau}(h) + \log c, \\ -\langle(\log B)h, h\rangle &\leq -\int_{\Delta} q \log(q/c) d\nu = S_{\omega}(h) + \log c. \end{aligned}$$

Rumin’s monotone form-domain argument handles possible zero densities. The last three displays again give (CK), without sigma-finiteness. If $c = 0$, Parseval for the ω -frame would force every $\tau_{\alpha} = 0$, contradicting Parseval for the τ -frame on a nonzero space; hence the logarithm above always has $c > 0$. The real case follows by complexification.

Claim. *The Continuous Kraus Conjecture, Conjecture 2.5 of arXiv:2310.01450, is true for the stated measure spaces and 1-bounded continuous Parseval frames.*

Independent later confirmation. Theorem 2.4 on page 3 of Krishna [3] proves a continuous operator-valued-frame uncertainty inequality on finite measure spaces by Riesz–Thorin interpolation. Taking the coefficient space to be the scalar field and $A_{\alpha}x = \langle x, \tau_{\alpha} \rangle$, $B_{\beta}x = \langle x, \omega_{\beta} \rangle$ gives $\|B_{\beta}A_{\alpha}^*\| = |\langle\tau_{\alpha}, \omega_{\beta}\rangle|$ and hence the finite-measure case of (CK) verbatim.

Provenance and scope. Rumin’s theorem predates the conjecture, so its author did not explicitly know he was answering it. Krishna’s 2024 paper does not cite or name Conjecture 2.5. Thus the relation is an agent-identified implication and the packet belongs in `literature_implied_answers`. The entropy inequality is fully resolved; the separate equality-characterization Question 2.4 of [1] remains outside this packet.

Search evidence. The run’s four cheap indexes contained no record for arXiv:2310.01450. A bounded arXiv/web search used the phrases “Continuous Kraus Conjecture”, “continuous frames entropic uncertainty”, “POVM entropic uncertainty”, and the exact arXiv id. It found Rumin’s earlier theorem and Krishna’s 2024 finite-measure theorem, but no paper explicitly naming Conjecture 2.5 as resolved.

References

- [1] K. Mahesh Krishna, *Continuous Deutsch Uncertainty Principle and Continuous Kraus Conjecture*, arXiv:2310.01450 (2023).
- [2] M. Rumin, *An entropic uncertainty principle for positive operator valued measures*, arXiv:1109.5889v2 (2011).
- [3] K. Mahesh Krishna, *Continuous Krishna–Parthasarathy Entropic Uncertainty Principle*, arXiv:2405.08003 (2024), published in *Infinite Dimensional Analysis, Quantum Probability and Related Topics*, doi:10.1142/S0219025724400022.